METRIC

MIL-I-62676
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MILITARY SPECIFICATION

INDICATORS AND TRANSMITTERS: LIQUID, TEMPERATURE AND PRESSURE (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers liquid quantity, pressure, and temperature indicators and transmitters for 24 volt (nominal) direct current (dc) electrical systems (see 6.1). The transmitters are of the electrical resistance type.

1.2 Classification. Indicators and transmitters shall be of the following units, types and classes, as specified (see 6.2):

Unit 1	- Indicators.
Type I	- Liquid quantity.
Type II	- Temperature.
Type III	- Pressure.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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Unit 2 - Transmitters.
Type I - Liquid quantity.
Class 1 - Float type.
Class 2 - Submerged tube type.
Type II - Temperature.
Type III - Pressure.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. The following specification, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

FEDERAL

L-P-380	- Plastic Molding Material, Methacrylate.
QQ-S-571	- Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.

MILITARY

MIL-W-80	- Window, Observation, Acrylic Base, Antielectrostatic, Transparent (for Indicating Instrument).
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.
MIL-P-81390	- Plastic, Molding Material, Polycarbonate, Glass Fiber Reinforced.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-595	- Colors used in Government Procurement.

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MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Navy Publication and Printing Services Office, Standardization Document Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

ARMY

12369000	- Chemical Agent Resistant Coatings (CARC).

(Unless otherwise indicated, U.S. Army drawings are available from U.S. Army Tank-Automotive Command, ATTN: Contracting Officer, Warren, MI 48397-5000.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3935	- Standard Specification for Polycarbonate (PC) Unfilled and Reinforced Material.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Materials. Materials shall be as specified herein and in applicable specifications, standards, or drawings. Materials not specifically designated shall be suitable for use in indicators and transmitters operating over the ranges specified without any change in physical or dimensional properties that would result in operation of the units falling outside of the limits specified herein (see 4.7.1).

3.2.1 Radioactivity. Indicators furnished under this specification shall not contain radioactive material (see 4.7.1).

3.2.2 Solder. Solder employed in making electrical connections shall conform to QQ-S-571 and a rosin flux shall be used (see 4.7.1).

3.2.3 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.7.1 and 6.4.3).

3.3 Design and construction. Indicators and transmitters shall be constructed to the form and dimensions specified in the applicable standards or drawings (see 4.7.1, 4.7.2, and 6.2).

3.3.1 Screw threads. All screw threads shall be in the metric system and shall conform to FED-STD-H28 (see 4.7.1).

3.3.2 Unit 1 indicators.

3.3.2.1 Dial.

3.3.2.1.1 Dial marking. The face of the dial shall be marked as shown in the applicable standard or drawing. Dial markings not covered by MIL-STD-130, such as, numerals, letters, and graduation marks shall have a minimum height of 3 millimeters (mm). Numerals, letters, and graduations shall be white conforming to color 37875 of FED-STD-595. Unless otherwise specified (see 6.2), numerals, graduations, and other markings shall be clearly defined in white on a black background (see 4.7.1 and 4.7.2).

3.3.2.1.2 Fastening. The dial shall be secured to the body of the indicator in such a position that the marking shall be legible when illuminated through the case windows from an outside source. The method of securing the dial shall be such that it will not become loose or turn when subject to vibration or shock (see 4.7.4.1.4 and 4.7.4.2).

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3.3.2.1.3 Pointer. The pointer shall be attached to the indicator movement and shall be damped to reduce oscillation from vibration. The pointer shall not touch the dial. The shape of the pointer shall not interfere with reading any point in the range of the dial. At least half of the visible portion of the pointer shall be white conforming to color 37875 of FED-STD-595 (see 4.7.2).

3.3.2.1.4 Cover window. The cover window shall be treated on both sides with an electrostatic solution, unless the window material has been tested for static charge generating characteristics and has been demonstrated to be nonsusceptible to a static charge that would deflect the pointer when the window is rubbed with a cloth containing not less than 95 percent fleece or pulled wool. The window shall be secured to the base so that moisture is excluded from the instrument case. The thickness of the window shall be 3 mm minimum. Window shall have an izod strength of not less than 0.4 Newton Meters (N·m) and shall retain transparency with age. The window material shall conform to MIL-W-80 or, if molded, shall conform to type I, class 3 of L-P-380 or Group 1, Class 1 of ASTM D3935 (see 4.7.2).

3.3.2.2 Case. The case shall be constructed of steel suitable for the purpose or plastic molding material conforming to Group 1, Class 1 of ASTM D3935 or MIL-P-81390 (see 4.7.1).

3.3.3 Unit 2 transmitters.

3.3.3.1 Mounting.

3.3.3.1.1 Type I. Type I transmitters shall be provided with a flange to permit attachment to fuel tanks. The diameters of the flange and boltholes and bolthole spacing shall be as specified in the applicable standard or drawing (see 4.7.1 and 4.7.2).

3.3.3.1.2 Types II and III. Types II and III transmitters shall be provided with a male standard pipe thread connection of the size specified on the applicable standard or drawing (see 4.7.2).

3.4 Performance. Unless otherwise specified, indicators and transmitters covered by this specification shall meet the performance requirements at all specified levels of temperature, pressure, and electrical input, as applicable (see 4.7.3).

3.4.1 Calibration.

3.4.1.1 Unit 1 indicators. Unless otherwise specified in the applicable standard or drawing, the centerline of the pointer shall indicate the centerline of the appropriate dial graduation within plus or minus three angular degrees. When current is off, the pointer shall return to the extreme left position (see 4.7.3.2).

3.4.1.2 Unit 2 transmitters. The transmitter resistance measurements shall be in accordance with the applicable standard or drawing (see 4.7.3.3, 4.7.3.4, or 4.7.3.5).

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3.4.2 Voltage variation: indicators. The indicators shall conform to 3.4.1.1 with an additional allowable tolerance of plus two angular degrees, minus four angular degrees, when first subjected to a 20 volt direct current (vdc) and then to a 30 vdc power source (see 4.7.3.6).

3.4.3 Overload capacity: transmitters.

3.4.3.1 Type II. Type II transmitters shall conform to 3.4.1.2 and shall evidence no mechanical or electrical damage after exposure for one minute to a temperature of 177 degrees Celsius (°C) (see 4.7.3.7.1).

3.4.3.2 Type III. Type III transmitters shall withstand a pressure of 200 percent of the maximum pressure specified in the applicable standard or drawing without evidence of mechanical or electrical damage. After this test, the transmitter shall conform to the requirements of 3.4.1.2 (see 4.7.3.7.2).

3.4.4 Attitude error: indicators. Indicators shall conform to 3.4.1.1 with an additional allowable tolerance of plus or minus two angular degrees when positioned and mounted, as follows (see 4.7.3.8):

a. 45 degrees clockwise from normal (vertical plane).
b. 45 degrees counterclockwise from normal (vertical plane).
c. Face up (horizontal plane).
d. Face down (horizontal plane).

3.4.5 Reverse connection capability: indicators (unit 1). The indicators shall evidence no damage and shall conform to the requirements of 3.4.1.1 with an additional allowable tolerance of plus or minus one angular degree after being subjected to a reverse in polarity from a 28 ± 0.5 vdc power source for a period of 2 hours minimum (see 4.7.3.9).

3.4.6 Dielectric strength. The insulation of indicators and transmitters with metal cases only shall show no evidence of cracking or charring when subjected to a potential of 400 volts root mean square (rms) at a frequency of 60 cycles per second for one minute (see 4.7.3.10).

3.4.7 Vibration resistance. Type I transmitters shall be tested in accordance with 4.7.4.1.1. Type II transmitters shall be tested in accordance with 4.7.4.1.2. Type III transmitters shall be tested in accordance with 4.7.4.1.3. Indicators shall be tested in accordance with 4.7.4.1.4. While indicators and transmitters are being vibrated, instruments shall be monitored to determine lapse of operation. Lapse of operation shall constitute test failure. Subsequent to the test, measurements shall be taken to determine whether the items meet the calibration requirements of 3.4.1 with added tolerance margins specified in 3.4.14. The items shall evidence no mechanical or electrical damage throughout the test.

3.4.8 Shock resistance. Types I, II, and III indicators and types II and III transmitters shall be tested as specified in 4.7.4.2. After testing, indicators and transmitters shall evidence no mechanical or electrical damage and shall meet the calibration requirements of 3.4.1 with added tolerance margins as follows:

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a. Types I, II and III indicators shall have an additional tolerance of plus or minus two angular degrees.
b. Type II transmitters shall have an additional tolerance of plus or minus one per cent of specified resistance at each calibration point.
c. Type III transmitters shall have an additional tolerance of plus 2.0 ohms at zero and plus or minus 1.0 ohm at the other checkpoints specified in the applicable standard or drawing.

3.4.9 Temperature influence.

3.4.9.1 Indicators. The indicators shall operate throughout an ambient air temperature range of minus 54 to plus 71°C and shall conform to 3.4.1.1 with an additional allowable tolerance of plus or minus five angular degrees (see 4.7.4.3.1).

3.4.9.2 Transmitters. The transmitters shall operate throughout an ambient air temperature of minus 54 to plus 71°C for type I, and minus 54 to plus 127°C for type III and shall conform to 3.4.1.2 with an additional allowable tolerance of plus 1.0 ohm at empty or zero and plus or minus 1.0 ohm at the other checkpoints specified in the applicable standard or drawing (see 4.7.4.3.2).

3.4.10 Waterproofness. Unit 1 indicators and unit 2 transmitters shall be waterproof when submerged and shall evidence no adverse effects on their mechanical and electrical capabilities (see 4.7.4.4 through 4.7.4.4.3, as applicable).

3.4.10.1 Types I, II, and III indicators and types II and III transmitters. Types I, II, and III indicators and types II and III transmitters shall be waterproof and shall be mechanically and electrically operable during and after submergence. There shall be no evidence of leakage while submerged (see 4.7.4.4.1 and 4.7.4.4.2).

3.4.10.1.1 Type III transmitters. Type III transmitters shall also be capable of sustaining a pressure of 124 kilopascals (kPa) for 30 minutes while submerged without showing any evidence of leakage. After submergence, the type III transmitter shall meet the requirements of 3.4.9.2 for all calibration points specified in the applicable standard or drawing (see 4.7.4.4.1.1).

3.4.10.2 Type I transmitter. Type I transmitter components including floats, tubes, and covers shall show no evidence of leakage while submerged in a saline solution at a temperature of 93°C minimum. Covers shall show no evidence of leakage under a pressure of 41 kPa with the cover mounted on the tank (see 4.7.4.4.3).

3.4.11 Corrosion resistance. The indicators and transmitters shall evidence no corrosion affecting operation after being subjected to a saline atmosphere of 200 hours minimum. Subsequently, the items shall meet the calibration requirements of 3.4.1 and shall show no evidence of mechanical or electrical damage (see 4.7.4.5).

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3.4.12 Fungus resistance. The indicators and transmitters shall evidence no fungus growth affecting operation after being subjected to hot-humid (tropic) conditions for a minimum of 90 continuous days. Subsequently the items shall meet the calibration requirements of 3.4.1 and shall show no degree of mechanical or electrical damage (see 4.7.4.6).

3.4.13 Transient voltage. The indicators shall show no evidence of damage or loss of calibration due to transient voltages (positive or negative) when connected in an electrical circuit as shown in figure 1 and 2 and subjected to a minimum of five pulses. Subsequently, the items shall meet the requirements of 3.4.1 (see 4.7.4.7 and 4.7.4.8).

3.4.14 Endurance. After testing in accordance with 4.7.4.9, the indicators and transmitters shall meet the requirements of 3.4.1 with the following allowable tolerance variations:

a. Type I, II, and III indicators shall have an additional allowable tolerance of plus or minus two angular degrees.
b. Type I transmitters shall have an additional allowable tolerance of plus 2.0 ohms at empty and plus or minus 1.0 ohm at full.
c. Type II transmitters shall have an additional allowable tolerance of plus or minus one per cent of resistance specified at each calibration point.
d. Type III transmitters shall have an additional allowable tolerance of plus 0.5 ohm at zero indicator pressure and plus or minus 1.0 ohm at the other checkpoints specified in the applicable standard or drawing.

3.5 Surface treatment. Cleaning, pretreatment, priming and painting shall conform to an appropriate CARC paint system selected from Drawing no. 12369000. Color of paint top coat is optional except for the bezel (see 3.5.1 and 4.7.2).

3.5.1 Bezel finish. The color of the indicator bezel finish top coat shall be lusterless camouflage green, color number 383, 34094 per MIL-C-46168 or MIL-C-53039.

3.6 Marking. Marking for identification shall be in accordance with MIL-STD-130.

3.7 Workmanship. Workmanship shall be such as to assure a product free of rust, burrs, and scratches. All electrical connections shall be mechanically secured, before being soldered or brazed, to preclude the possibility of becoming loose due to vibration or shock. All solder cleaning agents and flux shall be noncorrosive. All traces of flux shall be chemically neutralized and removed after soldering.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance shall comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (QCI) (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following conditions:

a. Temperature: 25 ± 8°C.
b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select four samples of each type and class of unit produced under the production contract for first article inspection. First article samples shall be inspected as specified in table I. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply items that are fully representative of those inspected as a first article sample. Any changes or deviations of the production units from the first article sample shall be subject to the approval of the contracting officer.

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TABLE I. Classification of inspections.

Title	Requirement	Inspection	First Article	QCI		
				Examination	Tests	Control
Group A:						
Materials, design, and construction	3.2 thru 3.3	4.7.1	X			
Defects (see 4.6.2 and table IV)	3.3, 3.5, 3.6 and 3.7	4.7.2	X	X		
Group B (Performance):						
Calibration:						
Indicator (unit 1)	3.4.1.1	4.7.3.2	X			X
Transmitters (unit 2)						
Type I	3.4.1.2	4.7.3.3	X		X	X
Type II	3.4.1.2	4.7.3.4	X		X	X
Type III	3.4.1.2	4.7.3.5	X		X	X
Voltage variation, indicators	3.4.2	4.7.3.6	X		X	
Overload capacity, transmitters	3.4.3	4.7.3.7	X			
Attitude error	3.4.4	4.7.3.8	X			
Reverse connection capability, indicators	3.4.5	4.7.3.9	X			
Dielectric strength	3.4.6	4.7.3.10	X			
Group C (Environmental):						
Vibration resistance	3.4.7	4.7.4.1	X			X
Shock resistance	3.4.8	4.7.4.2	X			
Temperature influence	3.4.9	4.7.4.3	X			X
Waterproofness						
Indicators (unit 1)	3.4.10.1	4.7.4.4.1	X			X
		4.7.4.4.2			X	
Transmitters, Types II and III	3.4.10.1	4.7.4.4.1	X			
		4.7.4.4.2			X	
Transmitters, Type III	3.4.10.1.1	4.7.4.4.1.1	X			X
Transmitters, Type I	3.4.10.2	4.7.4.4.3	X		X	X
Corrosion resistance	3.4.11	4.7.4.5	X			
Fungus resistance	3.4.12	4.7.4.6	X			
Transient voltage	3.4.13	4.7.4.7 and 4.7.4.8	X			
Group D (Life):						
Endurance	3.4.14	4.7.4.9	X			

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4.4.1 First article test sequence. First article tests shall be conducted on the sample in accordance with the test sequence specified in table II.

Table II. Order of first article testing.

UNIT 1 - INDICATORS	
Sample A: 4.7.3.2 4.7.3.6 4.7.4.3.1 4.7.4.9.1	- Calibration. - Voltage variation. - Temperature influence. - Endurance.
Sample B: 4.7.3.2 4.7.3.8 4.7.4.2 4.7.4.4.1 4.7.4.7 4.7.4.8	- Calibration. - Attitude error. - Shock. - Waterproofness. - Positive transient voltage. - Negative transient voltage.
Sample C: 4.7.3.10 4.7.3.2 4.7.4.1.4 4.7.4.5 4.7.3.9	- Dielectric strength. - Calibration. - Vibration. - Corrosion resistance. - Reverse connection capability.
Sample D: 4.7.3.2 4.7.4.6 4.7.3.2	- Calibration. - Fungus resistance. - Calibration.
UNIT 2 - TRANSMITTERS	
Sample A: 4.7.3.3, 4.7.3.4 or 4.7.3.5 4.7.4.3.2 4.7.4.9.2	- Calibration. - Temperature influence. - Endurance.
Sample B: 4.7.3.10 4.7.3.3, 4.7.3.4 or 4.7.3.5 4.7.4.1.1 (Type I) 4.7.4.1.2 (Type II) 4.7.4.1.3 (Type III) 4.7.4.5	- Dielectric strength. - Calibration. - Vibration. - Vibration. - Vibration. - Corrosion resistance.

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Table II. Order of first article testing - Continued.

Sample C:	
4.7.3.3, 4.7.3.4 or 4.7.3.5	- Calibration.
4.7.4.2	- Shock.
4.7.4.4.1 (Types II and III)	- Waterproofness.
4.7.4.4.1.1 (Type III)	- Waterproofness.
4.7.4.4.3 (Type I)	- Waterproofness.
4.7.3.7	- Overload.
Sample D:	
4.7.3.3, 4.7.3.4 or 4.7.3.5	- Calibration.
4.7.4.6	- Fungus resistance.
4.7.3.3, 4.7.3.4 or 4.7.3.5	- Calibration.

4.4.2 First article inspection failure. Deficiencies found during, or as a result of, the first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article inspection shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article inspection is completed to the satisfaction of the Government.

4.5 QCI. QCI shall include the examination of 4.5.2 and the tests of 4.5.3. Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection of the sample and the inspection lot.

4.5.1 Sampling plan. Unless otherwise specified (see 6.2), the sampling plan specified herein shall be used. See 6.4.2 for definitions of sampling inspection terms.

4.5.1.1 Lot formation. An inspection lot shall consist of all indicators or transmitters of a single type, class, and part identification number (PIN), from an identifiable production period, from one manufacturer, from one manufacturing location, submitted at the same time for acceptance.

4.5.1.2 Sample. The sample for QCI examination and tests shall be randomly selected from the inspection lot in accordance with table III.

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TABLE III. Sampling plan for QCI.

QCI sampling plan			
Inspection lot size	Sample size		
	Examination		Test
	Major	Minor	
2 to 8	*	5	2
9 to 15	13	5	2
16 to 25	13	5	2
26 to 50	13	5	5
51 to 90	13	7	5
91 to 150	13	11	6
151 to 280	20	13	7
281 to 500	29	16	9
501 to 1200	34	19	11
1201 to 3200	42	23	13
3,201 to 10,000	50	29	15
10,001 to 35,000	60	35	15
35,001 to 150,000	74	40	15
150,001 to 500,000	90	40	15
500,001 and over	102	40	15

* Indicates entire lot must be inspected (100% inspection).

4.5.2 Examination. The sample selected in accordance with 4.5.1.2 shall be examined and defects classified as specified in table IV (see 4.7.2). The acceptance number in all cases is zero.

TABLE IV. Classification of defects.

Categories	Defect	Method of inspection
Major:		
101	Dimensions affecting interchangeability not within tolerance (see 3.3).	Gage
102	Indicator dial markings do not conform to 3.3.2.1.1.	Visual
103	Indicator pointer does not conform to 3.3.2.1.3.	Visual
104	Indicator cover window does not conform to 3.3.2.1.4.	Visual
105	Type I flange dimensions and bolt hole spacings do not conform to applicable standard or drawing (see 3.3.3.1.1).	Visual

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TABLE IV. Classification of defects - Continued.

Categories	Defect	Method of inspection
106	Type II and III threaded connections not of size specified on applicable standard or drawing (see 3.3.3.1.2).	Gage
Minor:		
201	Dimensions not affecting interchangeability not within tolerance (see 3.3).	Gage
202	Defective surface treatment (see 3.5).	Visual
203	Improper marking (see 3.6).	Visual
204	Faulty workmanship (see 3.7).	Visual

4.5.3 Tests. The sample selected in accordance with 4.5.1.2 shall be subjected to the quality conformance tests specified in table I. The acceptance number in all cases is zero.

4.5.4 QCI failure. Any item that fails to conform to any specified requirement shall be rejected; any failure (one or more) of the selected sample in either the Major/Minor categories or test for the appropriate inspection lot size shall constitute a failure of the entire lot. The rejected item(s) may be repaired or corrected and resubmitted for inspection. If the contractor utilizes sampling inspection as an element of his inspection system, rejected inspection lots may be resubmitted for acceptance if the contractor performs 100 percent inspection on the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units or obtains procuring activity approval to resample the lot due to the insignificance of the defects. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots.

4.6 Control tests. Control tests shall be conducted on three items from each lot of 500 items consecutively produced of a particular unit, type, and class, except that not less than three nor more than six items shall be selected in any 30-day period. The selected items shall then be subjected to the examination of 4.5.2 and to a separate series of tests as follows (see table I):

Item 1:	Calibration	4.7.3.2, 4.7.3.3, 4.7.3.4 or 4.7.3.5
	Vibration	4.7.4.1
Item 2:	Calibration	4.7.3.2, 4.7.3.3, 4.7.3.4 or 4.7.3.5
	Waterproofness	4.7.4.4.1 or 4.7.4.4.3
	Transient voltage	4.7.4.7 and 4.7.4.8

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Item 3:	Calibration	4.7.3.2, 4.7.3.3, 4.7.3.4 or 4.7.3.5
	Waterproofness	4.7.4.4.1 or 4.7.4.4.3
	Temperature influence	4.7.4.3

4.6.1 Failure. Failure of any item to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.7 Methods of inspection.

4.7.1 Materials, design and construction. Conformance to 3.2 and 3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3, 3.3.2.1.1, 3.3.2.1.3, 3.3.2.1.4, 3.3.3.1.1, 3.3.3.1.2, 3.5, 3.6 and 3.7 shall be determined by examination for defects listed in table IV (see 4.5.2). Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.7.3 Performance tests.

4.7.3.1 Conditions. Items to be tested shall first be stabilized for one hour under the inspection conditions specified in 4.3 and shall be conducted in a laboratory free of dust, drafts, fumes, and products of combustion under the inspection condition specified in 4.3. Readings shall be taken on indicators and from type II transmitters only after voltage has been applied for a minimum of five minutes. Indicators and type I and type III transmitters may be lightly tapped while readings are being taken.

4.7.3.2 Calibration: indicators (unit 1).

4.7.3.2.1 Apparatus. Apparatus shall include a test chamber with air temperature variable within a range from minus 54°C to plus 71°C, a protractor, a voltmeter, an ammeter, a decade resistance box, and a source of 28 ± 0.5 volt electrical energy.

4.7.3.2.1.1 Calibration of test equipment. Test equipment resolution shall be such as to permit measurement of 10 percent of product tolerance. Instruments such as voltmeters, ammeters, and thermometers shall be of the laboratory type and shall be accurate to within ± 0.50 percent. The protractor shall be accurate to within ± 0.25 degrees and shall be free from error due to parallax. After the test equipment's accuracy has been established, the test equipment shall be calibrated at regular intervals to maintain their required accuracy. The Government inspector may refuse acceptance of the production items where the accuracy of the test equipment cannot be verified to the Government's satisfaction.

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4.7.3.2.2 Procedure. To determine conformance to 3.4.1.1, the indicator shall be mounted in a vertical position. The temperature shall be held at 25 ± 8°C. With the unit terminals connected to the voltage source and with resistance applied as specified in the applicable standard or drawing, dial pointer readings shall be taken by means of a precision protractor at each calibration point.

4.7.3.3 Calibration: type I transmitter (unit 2).

4.7.3.3.1 Apparatus. Apparatus shall include a fuel tank equipped with a fixture for attaching the unit, a Wheatstone bridge or ohmmeter, and test liquid having a specific gravity of .721 ± 0.0036.

4.7.3.3.2 Procedure. To determine conformance to 3.4.1.2, the following procedure shall apply. Calibration shall be on a descending scale. With the tank filled with liquid to such a level that the float (class 1) or the tube (class 2) is at the full position, the transmitter shall be attached to the fixture installed in the tank. The resistance of the item shall then be measured by means of a Wheatstone bridge or ohmmeter. The test shall be repeated with the tank empty. Class 1 (float type) transmitters may be tested with the float at "full" and "empty" positions without being installed in the fuel tank.

4.7.3.4 Calibration: type II transmitter (unit 2).

4.7.3.4.1 Apparatus. Apparatus shall include an oil tank provided with means for controlling the oil temperature and recirculating the oil, a laboratory type milliammeter, a laboratory type voltmeter, a supply of engine lubricating oil, and a source of 28 ± 0.5 volt dc electrical energy.

4.7.3.4.2 Procedure. To determine conformance to 3.4.1.2, the following procedure shall apply. Calibration for type II transmitters shall be on an ascending scale. The item shall be immersed in a tank filled with oil, starting at the lowest temperature specified in the applicable standard or drawing. The oil shall then be circulated at a minimum rate of 3.0 meters per minute past the item. The item shall subsequently be subjected to each of the temperatures specified in the applicable standard or drawing and the internal resistance at each temperature determined.

4.7.3.5 Calibration: type III transmitter (unit 2).

4.7.3.5.1 Apparatus. Apparatus shall include a dead weight tester or an oil or air tank with a provision for controlling the pressure, a precision pressure gage, and a Wheatstone bridge or ohmmeter.

4.7.3.5.2 Procedure. To determine conformance to 3.4.1.2, measurements shall be recorded in accordance with following procedure. Calibration shall be on an ascending scale, starting at the lowest pressure. The unit shall be subjected to each of the pressure values specified in the applicable standard or drawing for one minute and the internal resistance measured with the Wheatstone bridge or ohmmeter.

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4.7.3.6 Voltage variation. To determine conformance to 3.4.2, the indicator shall be subjected to the test specified in 4.7.3.2, except that the unit shall be connected first to a 20 volt dc and then to a 32 volt dc test circuit. Readings shall be taken at each value and conformance to 3.4.1.1 verified.

4.7.3.7 Overload capacity: transmitters (unit 2).

4.7.3.7.1 Type II transmitter. To determine conformance to 3.4.3.1, the type II transmitter shall be tested as specified in 4.7.3.4 except that it shall be held in a 177°C oil bath for one minute prior to testing. After testing, the item shall be examined for any evidence of mechanical or electrical damage.

4.7.3.7.2 Type III transmitter. To determine conformance to 3.4.3.2, the type III transmitter shall be subjected to an oil or air pressure of 200 per cent of that pressure specified for full scale reading on the applicable indicator for one minute and then tested as specified in 4.7.3.5. After testing, the item shall be examined for any evidence of mechanical or electrical damage.

4.7.3.8 Attitude error. To determine conformance to 3.4.4, the indicator shall be mounted in such a manner that the dial face can be positioned as follows:

a. 45 degrees clockwise from normal (vertical plane).
b. 45 degrees counterclockwise from normal (vertical plane).
c. Face up (horizontal plane).
d. Face down (horizontal plane).

With the exception of the mounting position, the procedure specified in 4.7.3.2.2 shall be followed. Dial pointer readings shall be taken for each of the four dial face positions and conformance to 3.4.1.1 verified.

4.7.3.9 Reverse connection capability: indicators (unit 1). To determine conformance to 3.4.5, the power source and signal connections to indicators shall be reversed for a period of 2 hours. A 28 ± 0.5 volt dc source shall be connected to the signal terminal and the lowest resistance specified in the applicable standard or drawing shall be applied to the power terminal on the indicator. Conformance to 3.4.1.1 shall be verified.

4.7.3.10 Dielectric strength for indicators and transmitters with metal cases only. To determine conformance to 3.4.6 and with the lead wires within the indicator or transmitter cases disconnected from ground or ground terminal, the ungrounded winding or circuit shall be subjected to a potential of 440 volts rms at a frequency of 60 cycles per second (cps) applied between the case and ungrounded terminal or winding for one minute. For sealed items, this test shall be conducted prior to final assembly. After testing, the insulation shall be examined for any evidence of cracking or charring.

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4.7.4 Environmental and life tests.

4.7.4.1 Vibration (units 1 and 2). For vibration testing, the transmitter and applicable indicator shall be mounted in a test fixture simulating actual mounting in use and both items tested simultaneously (see 3.4.7).

4.7.4.1.1 Vibration: type I transmitter (unit 2). To determine conformance to 3.4.7, type I transmitters shall be vibrated in the direction of each of the three major axes as specified in method 201 of MIL-STD-202. At the end of each vibration period along each of the three major axes and while the items are still being vibrated, the items shall be subjected to the test specified in 4.7.3.3. No evidence of mechanical or electrical damage shall occur throughout the test.

4.7.4.1.2 Vibration type II transmitter. To determine conformance to 3.4.7, type II transmitters shall be connected in a series circuit with a 28 ± 0.5 volt dc power source and applicable indicator. Care shall be taken to see that the mounting is free from resonance over the frequency range. While energized, the transmitter shall be subjected to simple harmonic motion having an amplitude of 0.76 mm or 50 gravity units (G) peak, whichever is less. Tolerance of plus or minus ten percent is permissible for the amplitude. The vibrational frequency shall be varied over the range from 10 to 3600 cps. Rate of frequency change shall be logarithmic. When there is no provision for logarithmic cycling, other automatic cycling rates of frequency change may be used. The vibrational cycle from 10 to 3600 and back to 10 cps shall be accomplished in 20 ± 2 minutes. This scanning cycle shall be repeated three times in order that critical frequencies (see 6.4) may be identified, recorded, and checked. After the scanning cycles, the transmitter shall be vibrated at the critical frequency for two hours. If there is more than one critical frequency, the two-hour period shall be divided equally between the critical frequencies. If there are more than three critical frequencies, the three most critical shall be selected and the transmitter vibrated for 40 minutes at each of the frequencies selected. If no critical frequency is identified, the item shall be vibrated at 50G acceleration with the frequency cycled from 1400 to 3400 and back to 1400 cps. Rate of change of frequency shall be logarithmic or, where there is no provision for logarithmic cycling, other automatic cycling rates of frequency change may be used. Time for each cycle shall be 20 ± 2 minutes. Duration of the test shall be two hours (six complete cycles). At the end of the two hour vibrational period, the transmitter shall be subjected to the test specified in 4.7.3.4. This test procedure shall be accomplished along each of three mutually perpendicular axes. Total vibrating time shall be nine hours (three hours along each axis). No evidence of mechanical or electrical damage shall occur throughout the test.

4.7.4.1.3 Vibration: type III transmitter. To determine conformance to 3.4.7, the transmitter shall be tested in accordance with test condition A, method 204 of MIL-STD-202. During the test, the transmitter shall be rigidly mounted as in normal operating position and shall be connected in a series circuit with a 28 ± 0.5 volt dc power source and applicable indicator. No evidence of mechanical or electrical damage shall occur throughout the test.

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4.7.4.1.4 Vibration: indicators. To determine conformance to 3.4.7, the indicator shall be mounted vertically and vibrated in each of the three major axes as specified in method 201 of MIL-STD-202. At the end of each vibration period the indicator shall be subjected to the test specified in 4.7.3.2. No evidence of mechanical or electrical damage shall occur throughout the test.

4.7.4.2 Shock. To determine conformance to 3.4.8, the items indicated shall be subjected to the applicable tests specified in 4.7.3.2, 4.7.3.4, or 4.7.3.5, after having been subjected to the following test. Types I, II, and III indicators and types II and III transmitters shall be mounted to simulate actual mounting in use. The items shall be subjected to the shock test as specified in method 213 of MIL-STD-202 in such manner to obtain a shock pulse of 1000 g's. The test shall be repeated 10 times and applied on the vertical axis while the test item is in its normal mounting orientation. After testing, there shall be no evidence of mechanical or electrical damage.

4.7.4.3 Temperature influence.

4.7.4.3.1 Unit 1, indicators. To determine conformance to 3.4.9.1, the indicator shall be subjected to the test specified in 4.7.3.2, except that the ambient air temperature shall be stabilized at minus 54 ± 3°C and subsequently at plus 71 ± 3°C. The indicator shall be soaked for four hours in the test chamber until its temperature is stabilized at that of the ambient air. Readings shall be recorded at each temperature.

4.7.4.3.2 Unit 2, transmitters. To determine conformance to 3.4.9.2, transmitters shall be subjected to the applicable test specified in 4.7.3.3, 4.7.3.4 or 4.7.3.5, except that the ambient temperature and the transmitter temperature shall be stabilized, and the resistance of the unit measured, at each of the following temperatures.

Type I	-	Plus 71 ± 3°C and minus 54 ± 3°C.
Type III	-	Plus 127 ± 3°C and minus 54 ± 3°C.

4.7.4.4 Waterproofness.

4.7.4.4.1 Waterproofness, first article and control tests. To determine conformance to 3.4.10.1, types I, II, and III indicators and types II and III transmitters shall be subjected to the waterproofness test, method 100 for type II, class 2 components specified in MIL-STD-1184.

4.7.4.4.1.1 Waterproofness, first article and control tests, type III transmitters. To determine conformance to 3.4.10.1.1, type III transmitters shall also be subjected to the waterproofness test, method 100, procedure 5 of MIL-STD-1184, except test pressure shall be 124 kilopascals (kPa) for 30 minutes. The threaded end shall be capped during the test. After testing, the type III transmitter shall be subjected to the test specified in 4.7.4.3.2 and conformance to 3.4.9.2 verified.

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4.7.4.4.2 Waterproofness, acceptance test. To determine conformance to 3.4.10.1, types I, II and III indicators and types II and III transmitters shall be subjected to an internal pressure of 41 kPa for 5 minutes while totally immersed in tap water. Observation for air bubbles from the interior of the component shall be maintained during the 5 minute period. Bubbles which are the result of entrapped air on the various exterior surfaces of the component shall not be considered a leak.

4.7.4.4.3 Waterproofness, type I transmitter. To determine conformance to 3.4.10.2, floats, tubes and covers on type I transmitters shall be tested for leakage by immersion in the saline solution of MIL-STD-1184 at a temperature of not less than 93°C. Covers shall be tested for leakage under a tank pressure of 41 kPa when the cover is mounted on the tank. Considerations relating to bubble formation of 4.7.4.4.2 shall apply.

4.7.4.5 Corrosion resistance. To determine conformance to 3.4.11, indicators and transmitters shall be subjected to the salt fog test specified in method 509 of MIL-STD-810, except that the test duration shall be 200 hours. When the corrosion test follows the vibration test, the additional tolerance margins specified in 3.4.7 shall be applicable. Subsequent to testing, indicators and transmitters shall meet the requirements of 3.4.1.1 or 3.4.1.2, as applicable.

4.7.4.6 Fungus resistance. To determine conformance to 3.4.12, indicators and transmitters shall be subjected to the fungus test specified in method 508 of MIL-STD-810, except that the duration of the test period shall be 90 continuous days. Subsequently, indicators and transmitters shall be tested in accordance with 4.7.3.2 through 4.7.3.5.2 as applicable, to verify conformance to 3.4.1. There shall be no evidence of mechanical or electrical damage due to fungus growth.

4.7.4.7 Positive transient voltage. To determine conformance to 3.4.13, the indicator shall be connected in a test circuit as shown in figure 1. Subject the indicator to five pulses by cycling switch S1 from "B" to "A". Switch shall remain in each position for three minutes. Subsequently, the indicator shall then be tested in accordance with 4.7.3.2 to verify conformance to 3.4.1.1. There shall be no evidence of mechanical or electrical damage to positive transient voltage.

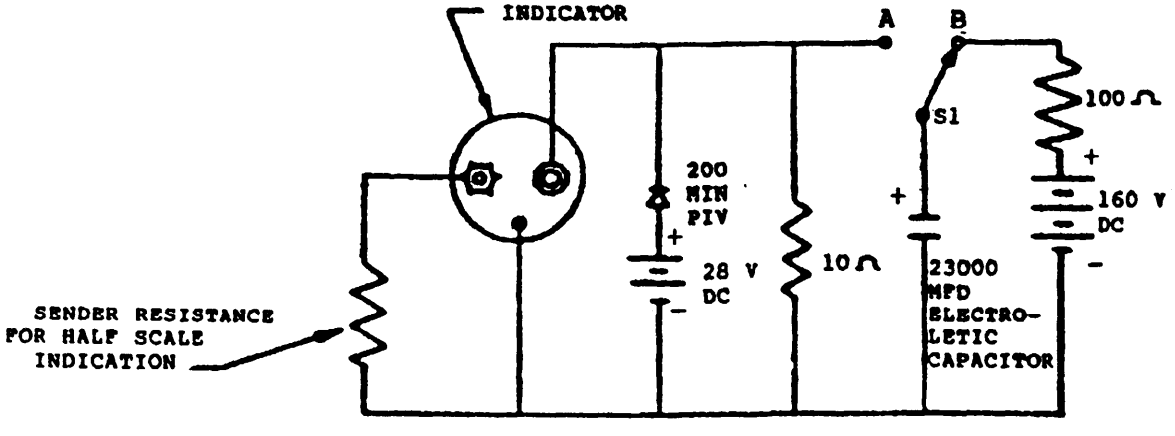


FIGURE 1. Positive transient test circuit.

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4.7.4.8 Negative transient voltage. To determine conformance to 3.4.13, the indicator shall be connected in a test circuit as shown in figure 2. Subject the indicator to five pulses by cycling switch S2 from "D" to "C". Switch shall remain in each position for one minute. Subsequently, the indicator shall then be tested in accordance with 4.7.3.2 to verify conformance to 3.4.1.1. There shall be no evidence of mechanical or electrical damage due to negative transient voltage.

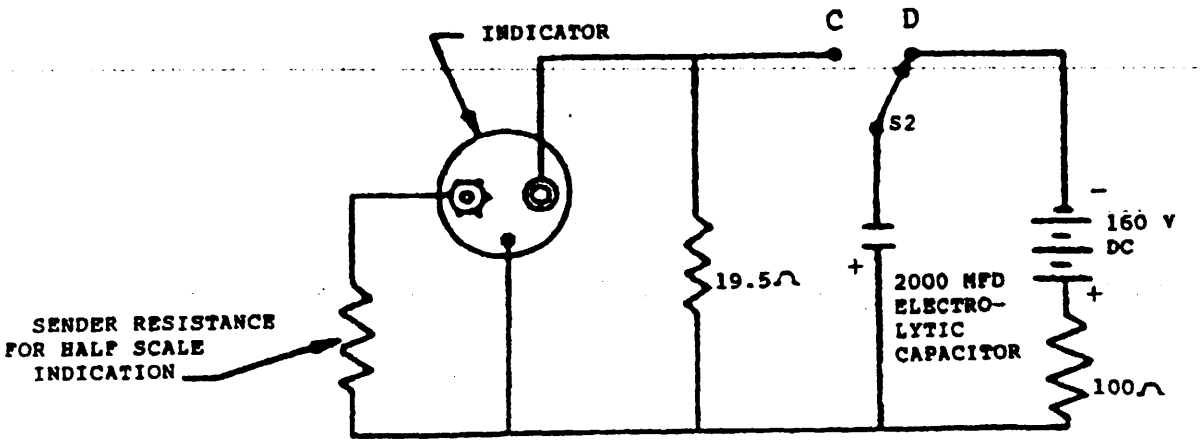


FIGURE 2. Negative transient test circuit.

4.7.4.9 Endurance.

4.7.4.9.1 Unit 1, indicators. To determine conformance to 3.4.14, the indicators shall be electrically connected to the applicable transmitter or an equivalent resistance and subjected to full scale deflection and return for a total of 200,000 cycles. One cycle of operation shall consist of one complete oscillation from extreme left to extreme right and return to extreme left. Cycling shall be at such rate as to produce full scale deflection of the indicator pointer. Subsequent to testing, the indicators shall be subjected to the test specified in 4.7.3.2 for conformance to 3.4.1.1.

4.7.4.9.2 Unit 2, transmitters. To determine conformance to 3.4.14, the transmitters shall be subjected to full scale deflection and return as indicated on the applicable indicator for a total of 100,000 cycles (10,000 cycles for type II transmitters). Cycling shall be at a rate such as to produce full scale deflection on the applicable indicator pointer. Transmitters may be tested concurrently with their applicable indicators. Subsequent to testing, the transmitters shall be subjected to the test specified in 4.7.3.3, 4.7.3.4, or 4.7.3.5, as applicable for conformance to 3.4.1.2.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2 and 6.4).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. An indicator (dash unit) and a transmitter (sending unit), when connected by a single conductor cable (not supplied with the device) and appropriately grounded, comprise a pressure, temperature, or liquid quantity gage. These devices are used in military motor vehicles and other internal combustion engine applications to inform the operator of the following: a) Type I: The level of fuel in the supply tank(s); b) type II: The engine or transmission coolant or other liquid temperature; and c) Type III: The oil pressure in the engine or transmission lubricating system or the air pressure in the airbrake system.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Unit, type, and class of item required (see 1.2).
c. Applicable standards or drawings (see 2.1 and 3.3).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
e. If first article samples are required (see 3.1).
f. Item name and drawing number (see 3.3).
g. If other than white markings on a black background are required (see 3.3.2.1.1).
h. If responsibility for inspection and the place of inspection is other than as specified (see 4.1).
i. If first article inspection is other than as specified (see 4.4).
j. Sampling plan for QCI inspection (see 4.5.1).
k. Selection of applicable level and packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers on whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Definitions.

6.4.1 Critical frequency. Critical frequency is a resonant frequency developed in the transmitter such that the amplitude exceeds that of the applied vibration. The greater the amplitude the more highly loaded the transmitter and the more critical the vibration environment.

6.4.2 Definitions of terms used in sampling inspection.

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirements. Defects will normally be grouped into one or more of the following classes: critical, major and minor defects. Also, defects may be grouped into other classes, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

c. Critical defective. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

d. Defective. A defective is a unit of product which contains one or more defects.

e. Formation of lots or batches. The product shall be assembled into identifiable lots, sublots, batches, or in such other manner as may be prescribed (see 1). Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time.

f. Lot or batch. The term lot or batch shall mean "inspection lot" or "inspection batch", i.e., a collection of units or product from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment, etc.).

g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

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i. Major defective. A major defective is a unit of product which contains one or more major defects, and may also contain minor defects but contains no critical defect.

j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

l. Presentation of lots or batches. The formation of the lots or batches, lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary, the supplier shall provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample shall be selected in proportion to the size of sublots or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch shall be selected at random.

n. Sample A sample consists of one or more units of product drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. The number of units or product in the sample is the sample size.

o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

p. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

6.4.3 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.4).

6.4.4 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows,

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or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.5 Cross-reference. Items conforming to this specification are not interchangeable/substitutable with items conforming to MIL-I-10986-G (English version).

6.6 Part identifying number (PIN). The following describes the PIN number construction for use with indicators and transmitters acquired under this specification.

M 62676-X-X-X

- Class (liquid quantity type I transmitters only):
  - 1 - Class 1 (Float type)
  - 2 - Class 2 (Submerged tube type)
- Type number (see 1.2):
  - A - type I (Liquid quantity)
  - B - type II (Temperature)
  - C - type III (Pressure)
- Unit:
  - 1 - Indicators
  - 2 - Transmitters
- Military specification number
- Prefix "M" indicates metric specification

6.7 Subject term (key word) listing.

Case
Corrosion resistance
Dial
Dielectric strength
Endurance
Fungus resistance
Mounting
Performance
Radioactivity
Shock resistance
Solder
Vibration resistance
Waterproofness
Workmanship

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6.8 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

Custodians:
Army - AT

Preparing activity:
Army - AT

(Project 6620-0499)

Review activities:
Army - ER
DLA - GS

User activities:
Army - ME
NAVY - MC

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